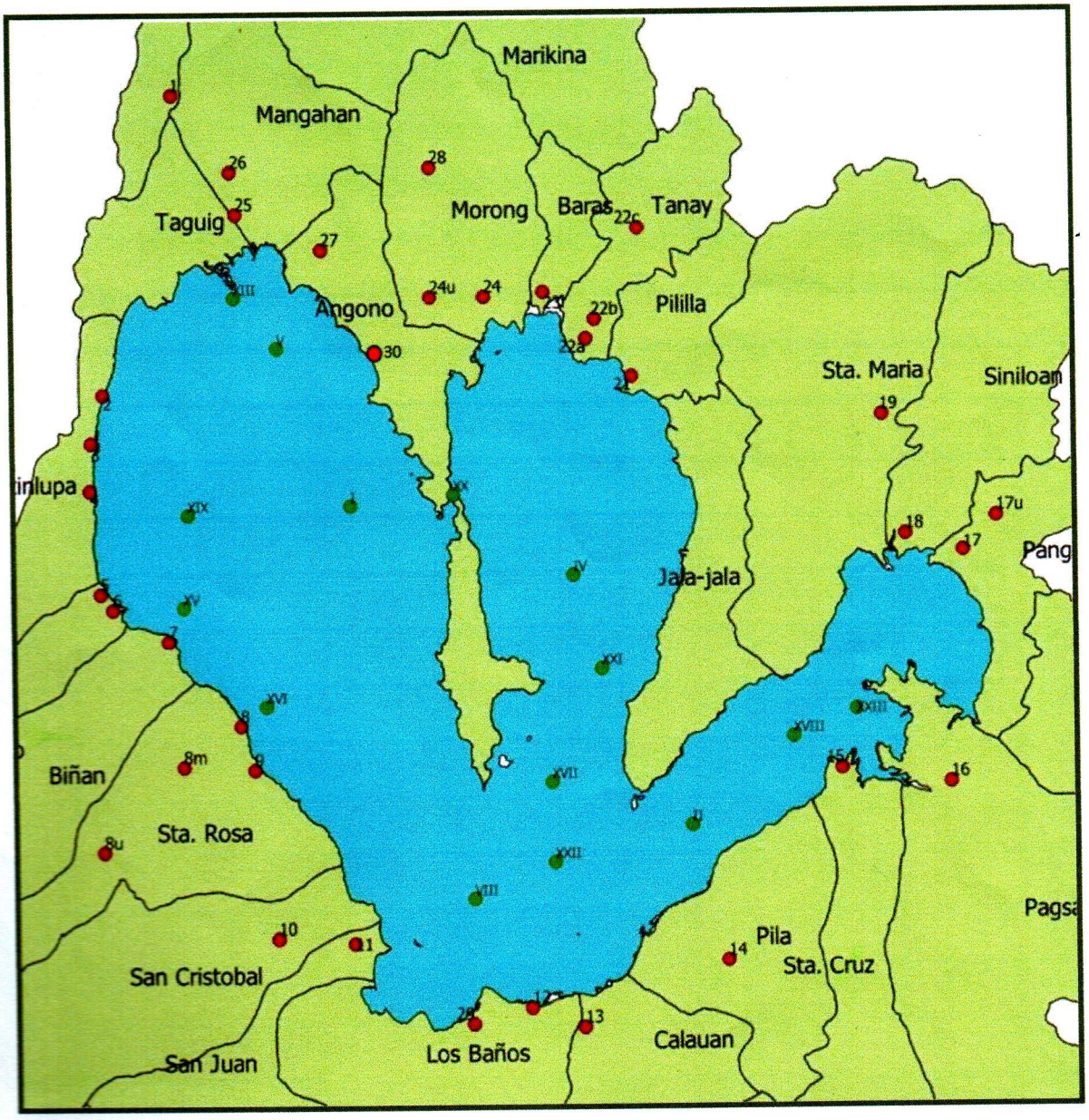
LLDA Quarterly Water Quality Monitoring Report
Laguna Lake and Tributary Rivers
July to September, 2023



Laguna Lake Development Authority
Resource Management and Development Department
Environmental Laboratory and Research Division

Quarterly Water Quality Report
Laguna Lake and Tributary Rivers
July to September, 2023

Laguna Lake Stations

Monitoring Stations		Water Quality Parameters								
		BOD (mg/L)			Dissolved Oxygen (mg/L)			Fecal Coliform, MPN/100mL (Geomean)		
		July	August	September	July	August	September	July	August	September
Stn. I	(Central West Bay)	2	2	3	6.7	6.4	7.6	41	146	146
Stn. II	(East Bay)	2	2	5	7.4	7.4	6.5	19	19	19
Stn. IV	(Central Bay)	<1	2	3	6.6	6.8	7.6	43	44	43
Stn. V	(Northern West Bay)	<1	2	3	6.7	7.6	7.2	19	26	39
Stn. VIII	(South Bay)	3	2	2	6.9	7.6	7.5	42	42	18
Stn. XIII	(Taytay)	2	3	2	7.2	7.4	6.8	57	32	19
Stn. XV	San Pedro (West Bay)	2	2	4	6.7	7.9	6.3	148	297	84
Stn. XVI	Sta Rosa (West Bay)	2	2	2	7.1	7.6	5.5	44	42	47
Stn. XVII	Fish Sanctuary (Central Bay)	2	2	1	7.5	7.6	7.3	63	63	63
Stn. XVIII	Pagsanjan (East Bay)	2	2	2	7.4	7.5	7.4	53	53	18
Stn. XIX	(Muntinlupa)	2	3	1	6.5	7.2	7.2	64	64	24
Stn. XX	(GEMS)	2	2	2	6.2	5.8	7.0	1,010	855	354
Stn. XXI	(Cardona)	3	2	2	7.5	7.4	8.1	49	77	126
Stn. XXII	(Jala-jala)	2	2	2	7.5	7.5	7.8	24	57	42
Stn. XXIII	(Lumban)	2	2	2	6.6	7.3	7.2	64	47	47

Tributary Rivers Stations

Station	Monitoring Stations	Water Quality Parameters								
		BOD (mg/L)			Dissolved Oxygen (mg/L)			Fecal Coliform, MPN/100mL (Geomean)		
		July	August	September	July	August	September	July	August	September
1	Marikina River		9	4		3.4	5.3		230,000	230,000
2	Bagumbayan River (Taguig)	125	299	146	<0.1	<0.1	<0.1	27,293,550	33,482,355	24,261,039
3	Buli Creek	79	127	1,202	<0.1	<0.1	<0.1	15,318,380	18,791,818	9,757,640
4	Mangangate River (Muntinlupa) -Downstream	18	26	35	<0.1	<0.1	<0.1	5,745,658	4,295,055	3,337,946
5	Tunasan River (Muntinlupa) -Downstream	24	141	68	<0.1	<0.1	<0.1	8,354,382	7,417,890	4,900,000
6	San Pedro River (T_2)	12	8	9	1.2	3.0	2.9	1,428,857	1,047,447	1,428,857
7	Biñan River	11	60	22	2.5	<0.1	0.7	4,288,848	4,485,297	3,914,019
8	Sta. Rosa River - Downstream	25	51	33	<0.1	0.3	0.5	3,764,795	8,271,860	14,047,458
8M	Sta. Rosa River - Midstream	36	38	26	1.3	0.9	3.1	3,300,000	3,764,795	3,764,795
8U	Sta. Rosa River - Upstream	5	8	6	6.6	6.6	6.2	446,530	332,795	642,841
9	Cabuyao River	18	12	52	<0.1	0.2	<0.1	6,372,013	5,945,903	5,945,903
10	San Cristobal River (T_3)		9	57	1.3	3.7	<0.1	24,818,959	13,508,088	46,781,847
11	San Juan River (T_1)		5	3	5.0	6.2	5.0	333,795	376,479	301,800
12	Molawin Creek (Los Baños)		1	3	5.7	5.8	5.0	125,571	203,400	273,996
13	Bay River (T_5)		3	2	5.8	7.2	5.0	59,055	114,416	210,222
14	Pila River		2	2	3.7	5.1	4.0	5,575	8,015	12,962
15	Sta. Cruz River (T_4)		1	3	5.6	5.9	6.0	45,688	36,625	28,249
16	Pagsanjan River (T_6)	<1	1	2	6.2	6.7	6.0	22,722	22,722	10,502
17	Pangil River - Downstream	<1	1	2	7.3	6.7	6.0	36,131	48,373	55,079
17U	Pangil River - Upstream	<1	1	2	7.6	7.5	6.0	4,870	4,141	3,338
18	Siniloan River	1	2	2	6.1	5.4	6.0	62,823	75,232	44,663
19	Sta. Maria River - Downstream	1	2	2	5.6	6.5	6.0	6,626	19,492	19,017
19U	Sta. Maria River - Upstream	<1	1	<1	7.8	7.8	6.0	4,295	5,793	5,946
20	Jala-jala River									
21	Pililla River		2	2	1.3	5.9	6.8	215,578	95,108	69,721
22 A	Tanay River - Brgy. Wawa (Downstream)		2	2	6.1	7.3	7.7	381,348	237,231	91,278
22 B	Tanay River - Midstream		<1	1	7.0	7.5	7.8	18,660	30,180	16,356
22 C	Tanay River - Daranak Falls (Upstream)		<1	1	7.7	7.6	7.9	13,391	13,064	9,991
23	Baras River		7	3	1.4	5.2	6.8	588,022	711,194	171,939
24	Morong River - Downstream	6	5	6	1.4	6.4	7.2	531,506	389,629	426,076
24U	Morong River - Upstream	3	3	3	3	4.9	5.6	24,469	33,874	38,206
25	Manggahan Floodway (Taytay)		4			1.3			889,944	
26	Sapang Baho River (Cainta)	14	6	4	<0.1	1.6	1.8	2,620,062	2,897,818	1,603,923
27	Angono River	5	5	2	3	5.5	6.3	1,817,373	606,367	426,076
28	Teresa River	3	5	5	2.1	6.0	6.6	84,743	64,671	92,819
29	Saran River									
30	Binangonan River	30	5	4	<0.1	6.0	1.7	62,698,992	29,102,294	6,181,579

Based on DENR Administrative Order No. 2016-08

For BOD:
- Conformed with Class A (concentration of less than & up to 3 mg/L)
- Conformed with Class B (concentration of more than 3 mg/L & up to 5mg/L)
- Conformed with Class C (concentration of more than 5 mg/L & up to 7 mg/L)
- Conformed with Class D (concentration of more than 7 mg/L & up to 15 mg/L)
- Failed the Water Quality Guidelines for Classes A to D (more than 15 mg/L)
- No data

For DO:
- Conformed with Classes A, B and C (concentration of more than & equal to 5 mg/L)
- Conformed with Class D (concentration of more than & equal to 2 mg/L up to less than 5 mg/L)
- Failed the Water Quality Guidelines for Classes A to D (less than 2 mg/L)
- No data

Based on DENR Water Quality Guidelines No. 2021-19

For Fecal Coliform:
- Conformed with Class A (≤ 50 MPN/100mL)
- Conformed with Class B (more than 50 up to 100 MPN/100mL)
- Conformed with Class C (more than 100 up to 200 MPN/100mL)
- Conformed with Class D (more than 200 up to 400 MPN/100mL)
- Failed the Water Guidelines for Classes A to D (more than 400 MPN/100mL)
- No data

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Laguna Lake Stations

Monitoring Stations		Water Quality Parameters														
		pH (units)			Ammonia (mg/L)			Nitrate (mg/L)			Inorganic Phosphate (mg/L)			Total Suspended Solids (mg/L)		
		July	August	September	July	August	September	July	August	September	July	August	September	July	August	September
Stn. I	(Central West Bay)	8.2	8.1	8.5	0.02	0.10	0.08	<0.1	0.21	0.5	0.17	0.18	0.10	15	83	27
Stn. II	(East Bay)	8.5	8.0	8.0	<0.01	0.07	<0.01	<0.1	0.22	0.1	0.17	0.13	0.12	4	47	18
Stn. IV	(Central Bay)	8.5	8.2	8.5	<0.01	0.01	<0.01	<0.1	0.60	0.2	0.23	0.17	0.14	7	34	18
Stn. V	(Northern West Bay)	8.6	8.3	8.5	<0.01	0.02	<0.01	<0.1	0.46	0.1	0.21	0.14	0.13	42	91	13
Stn. VIII	(South Bay)	8.5	8.3	8.4	0.03	<0.01	<0.01	<0.1	0.11	0.1	0.15	0.15	0.12	58	17	16
Stn. XIII	(Taytay)	8.8	8.4	8.3	<0.01	<0.01	0.01	<0.1	0.29	0.1	0.19	0.15	0.14	41	85	3
Stn. XV	San Pedro (West Bay)	8.5	8.5	8.4	0.01	<0.01	0.03	<0.1	0.25	0.2	0.17	0.16	0.16	17	27	11
Stn. XVI	Sta Rosa (West Bay)	8.4	8.5	8.1	0.02	<0.01	<0.01	<0.1	0.13	0.1	0.13	0.14	0.15	24	20	10
Stn. XVII	Fish Sanctuary (Central Bay)	8.7	8.4	8.4	<0.01	<0.01	<0.01	<0.1	0.11	0.1	0.21	0.16	0.14	13	61	12
Stn. XVIII	Pagsanjan (East Bay)	8.3	8.1	8.4	<0.01	0.02	<0.01	<0.1	0.28	0.1	0.08	0.11	0.13	14	77	8
Stn. XIX	(Muntinlupa)	8.4	8.0	8.6	0.05	0.01	<0.01	<0.1	1.30	0.2	0.19	0.14	0.14	8	63	5
Stn. XX	(GEMS)	8.1	8.0	8.5	<0.01	<0.01	0.02	<0.1	1.06	0.4	0.13	0.13	0.13	8	76	15
Stn. XXI	(Cardona)	8.6	8.3	8.7	<0.01	<0.01	<0.01	<0.1	<0.1	0.2	0.21	0.14	0.13	6	16	12
Stn. XXII	(Jala-jala)	8.6	8.5	8.4	0.02	<0.01	<0.01	<0.1	0.11	0.1	0.17	0.14	0.13	7	25	9
Stn. XXIII	(Lumban)	7.7	8.1	8.1	0.04	<0.01	<0.01	<0.1	<0.1	<0.1	0.09	0.14	0.11	8	85	10

Legend:

Based on DENR Water Quality Guidelines No.2016-08 :

For pH:
- Conformed with Classes A and B (acceptable range = 6.5 to 8.5)
- Conformed with Class C (acceptable range = 6.5 to 9.0)
- Conformed with Class D (acceptable range = 6.0 to 9.0)
- Failed the water quality guidelines for Classes A to D (less than 6 and more than 9)

For Nitrate:
- Conformed with Classes A, B and C (concentration of less than and up to 7 mg/L)
- Conformed with Class D (concentration of more than 7 mg/L and up to 15 mg/L)
- Failed the water quality guidelines for Classes A to D (more than 15 mg/L)

For Total Suspended Solids:
- Conformed with Class A (concentration of less than and up to 50 mg/L)
- Conformed with Class B (concentration of more than to 50 mg/L and up to 65 mg/L)
- Conformed with Class C (concentration of more than to 65 mg/L and up to 80 mg/L)
- Conformed with Class D (concentration of more than 80 mg/L up to 110 mg/L)
- Failed the Water Quality Guidelines for Classes A to D (concentration of more than 110 mg/L)

Based on DENR Water Quality Guidelines No. 2021-19:

For Ammonia:
- Conformed with Classes A , B and C (concentration of less than and up to 0.06 mg/L)
- Conformed with Class D (concentration of more than 0.06 mg/L and up to 0.30mg /L)
- Failed the Water Quality Guidelines for Classes A to D (more than 0.30 mg/L)

Based on DENR Water Quality Guidelines No. 2021-19:

For Phosphate:
- Conformed with Classes A , B and C (concentration of less than and up to 0.025 mg/L)
- Conformed with Class D (concentration of more than 0.025 mg/L and up to 0.05 mg/L)
- Failed the Water Quality Guidelines for Classes A to D (concentration of more than 0.05 mg/L)

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Tributary River Stations

Station	Monitoring Stations	Water Quality Parameters											
		pH (units)			Ammonia (mg/L)			Nitrate (mg/L)			Inorganic Phosphate (mg/L)		
		July	August	September	July	August	September	July	August	September	July	August	September
1	Marikina River		7.4	7.4		2.75	2.0		0.88	0.90		0.13	0.13
2	Bagumbayan River (Taguig)	7.1	6.8	7.2	24.9	22.00	21.0	0.12	1.69	0.10	2.6	2.30	3.00
3	Buli Creek	7.4	7.4	7.0	21.8	27.75	13.0	<0.1	0.79	0.10	2.1	2.10	2.00
4	Mangangate River (Muntinlupa) -Downstream	7.5	7.6	7.5	13.8	22.00	17.0	<0.1	0.48	<0.1	1.47	1.97	2.00
5	Tunasan River (Muntinlupa) -Downstream	8.6	7.5	7.3	8	9.25	6.00	0.13	0.84	0.11	1.96	1.92	1.00
6	San Pedro River (T_8)	7.5	7.7	7.8	4.4	4.70	6.00	1.92	1.25	2.00	0.95	1.42	3.20
7	Biñan River	7.4	7.6	7.6	2.6	19.50	9.00	1.16	0.13	0.18	0.5	2.25	4.10
8	Sta. Rosa River - Downstream	7.5	7.7	7.7	1.0	3.15	11.00	<0.1	<0.1	0.13	1.07	1.22	4.70
8M	Sta. Rosa River - Midstream	7.4	7.9	7.6	1.3	2.75	2.00	<0.1	0.12	0.14	0.41	0.62	3.10
8U	Sta. Rosa River - Upstream	7.9	8.0	8.0	1.29	2.11	2.00	1.51	2.17	1.00	0.52	0.86	0.70
9	Cabuyao River	7.5	7.5	7.5	8.6	7.60	5.00	<0.1	0.11	0.52	1.85	1.61	1.40
10	San Cristobal River (T_9)	7.5	7.2	7.1	<0.01	0.29	0.26	0.10	1.13	0.13	0.18	0.28	1.70
11	San Juan River (T_9)	7.4	7.5	7.6	0.58	0.36	0.37	3.02	3.27	4.00	0.92	0.60	2.10
12	Molawin Creek (Los Baños)	7.1	7.1	7.0	0.29	0.10	0.13	1.24	0.85	0.94	0.27	0.16	0.55
13	Bay River (T_9)	7.8	7.8	7.9	0.10	0.17	0.05	0.87	1.14	1.00	0.33	0.37	1.10
14	Pila River	7.2	7.2	7.3	0.08	0.08	0.05	0.47	0.56	0.56	0.127	0.16	0.60
15	Sta. Cruz River (T_6)	7.4	7.3	7.3	0.68	0.10	0.15	1.00	1.02	1.00	3.08	0.09	1.00
16	Pagsanjan River (T_9)	7.1	7.3	7.2	0.05	0.05	0.10	0.30	0.24	1.00	0.07	0.10	0.26
17	Pangil River - Downstream	7.2	7.5	7.2	0.05	0.05	0.07	0.21	0.22	0.90	0.08	0.10	0.32
17U	Pangil River - Upstream	7.0	7.1	7.1	<0.01	<0.01	0.03	<0.1	0.19	0.40	0.06	0.09	0.23
18	Siniloan River	6.7	6.7	6.7	0.47	0.13	0.20	0.45	0.44	2.00	0.07	0.09	0.23
19	Sta. Maria River - Downstream	7.0	7.2	7.2	0.31	0.04	0.10	0.49	0.42	1.00	0.05	0.09	0.27
19U	Sta. Maria River - Upstream	7.4	7.6	7.8	0.07	<0.01	0.03	0.29	0.35	0.80	0.10	0.09	0.31
20	Jala-jala River												
21	Pililla River	7.0	7.3	7.3	0.14	0.14	0.07	1.03	1.40	1.40	0.29	0.30	0.17
22 A	Tanay River - Brgy. Wawa (Downstream)	7.8	8.0	8.1	0.10	0.02	0.05	1.04	0.78	0.94	0.06	0.07	0.07
22 B	Tanay River - Midstream	7.7	8.3	8.2	0.03	0.01	0.03	0.92	0.74	0.91	0.05	0.06	0.07
22 C	Tanay River - Daranak Falls (Upstream)	8.0	8.2	8.1	0.01	0.12	0.06	0.73	0.78	0.77	0.05	0.08	0.07
23	Baras River	6.7	7.1	6.9	0.37	0.21	0.43	3.01	2.08	2.00	0.49	0.33	0.23
24	Morong River - Downstream	7.8	8.1	8.0	2.5	0.60	1.10	1.60	0.92	2.00	1.42	0.52	0.48
24U	Morong River - Upstream	7.7	7.8	7.7	7.1	0.61	1.20	0.83	1.53	2.00	1.69	0.55	0.43
25	Manggahan Floodway (Taytay)		7.2			2.50			0.47			0.26	
26	Sapang Baho River (Cainta)	7.4	7.2	7.2	9.37	1.95	1.50	0.19	0.52	3.00	1.18	0.34	0.31
27	Angono River	7.5	7.7	7.7	3.8	0.93	1.50	0.94	0.47	2.00	0.47	0.23	0.19
28	Teresa River	7.7	8.0	7.9	0.48	0.90	0.40	1.18	0.75	2.00	0.37	0.12	0.13
29	Saran River												
30	Binangonan River	7.3	7.5	7.3	12.7	0.35	1.70	0.19	0.55	1.00	1.39	0.56	0.72

Legend:
Based on DENR Water Quality Guidelines No.2016-08 :

For pH:
- Conformed with Classes A and B (acceptable range = 6.5 to 8.5)
- Conformed with Class C (acceptable range = 6.5 to 9.0)
- Conformed with Class D (acceptable range = 6.0 to 9.0)
- Failed the water quality guidelines for Classes A to D (less than 6 and more than 9)
- No data

For Nitrate:
- Conformed with Classes A, B and C (concentration of less than and up to 7 mg/L)
- Conformed with Class D (concentration of more than 7 and up to 15 mg/L)
- Failed the water quality guidelines for Classes A to D (more than 15 mg/L)
- No data

Based on DENR Water Quality Guidelines No. 2021-19

For Ammonia:
- Conformed with Classes A , B and C (concentration of less than and up to 0.06 mg/L)
- Conformed with Class D (concentration of more than 0.06 mg/L and up to 0.30mg /L)
- Failed the Water Quality Guidelines for Classes A to D (more than 0.30 mg/L)
- No data

For Phosphate:
- Conformed with Classes A , B and C (concentration of less than and up to 0.025 mg/L)
- Conformed with Class D (concentration of more than 0.025 mg/L and up to 0.05 mg/L)
- Failed the Water Quality Guidelines for Classes A to D (concentration of more than 0.05 mg/L)
- No data

FINDINGS FOR THE 3rd QUARTER OF 2023:

For the Tributary Rivers, no water samples were collected in the following monitoring tributary river stations:

Stn. 1 (Marikina River) in July - inaccessible due to the ongoing construction
Stn. 25 (Manggahan Floodway, Taytay) in July and September - prolific water hyacinth
Stn. 29 (Saran River) - sampling site was too shallow

*Note: Starting in June 2022, Stn. 20 (Jala-jala River) was discontinued and was replaced by Binangonan River as new sampling station.

A. On Biochemical Oxygen Demand (BOD):

Laguna Lake:

- The monthly concentrations in all monitoring stations consistently conformed with the Water Quality Guidelines (WQG) set by DENR for Class A except for Stn. II which was assessed as Class B in September.
- The highest monthly concentration was 5.0 mg/L recorded in Stn. II (East Bay) in September while the lowest was <1.0 mg/L in Stn. IV (Central Bay) and Stn. V (Northern West Bay) in July.

Tributary Rivers:

- Out of the thirty-six (36) monitored tributary river stations, the following **consistently** conformed with the WQG set for BOD in 3rd Quarter:
  - Class A - Seven (7) tributary river stations, namely; Stns. 16, 17, 17U, 18, 19, 19U, and 24U.
  - Class D – Only Stn. 6 (San Pedro River).
  - Failed the WQG for Classes A to D - Six (6) stations, specifically; Stns. 2, 3, 4, 5, 8, and 8M.
- The remaining tributary river stations were occasionally assessed for Class A, B, C, D, and even failed the WQG.
- The highest BOD level was recorded at 1,202 mg/L noted in Stn. 3 (Buli Creek) in September while the lowest was <1 mg/L in Stns. 16, 17, 17U, and 19U in July, Stns. 22B and 22C in August, and Stn. 19U in September.
- The sediment run-off in Buli Creek possibly caused the high spike in BOD which was was due to the aftermath of excessive heavy rains, in addition to the effects of domestic wastes, and the ongoing construction at the sampling location.

B. On Dissolved Oxygen (D.O.):

Laguna Lake:

- The monthly DO concentrations in fifteen (15) lake stations ranged from 5.5 mg/L to 8.1 mg/L, and all conformed with the WQG for Classes A, B and C.
- The highest and lowest monthly DO concentrations were noted in Stn. XXI (Cardona) and Stn. XVI (Sta. Rosa – West Bay), respectively, and were both recorded in September.

Tributary Rivers:

- Out of the thirty-six (36) monitored tributary river stations, the following consistently conformed with the WQG set for DO:
  - Classes A, B and C - Fourteen (14) tributaries, namely; Stns. 8U, 11, 12, 13, 15, 16, 17, 17U, 18, 19, 19U, 22A, 22B and 22C.
  - Failed the WQG for Classes A to D – Seven (7) monitored stations, namely; Stns. 2, 3, 4, 5, 8, 9, and 26.
- There were sporadic assessments of Class A, B, C, D, and those that failed to comply with the WQG in the remainder of the tributary rivers.
- The highest monthly concentration was at 8.0 mg/L in Stns. 17, 17U, and 19U which were all in September while the lowest was at was at <0.1 mg/L noted in Stns. 2, 3 4 and 5 in July to September; Stn. 7 in August; Stns. 8, 26 and 30 in July; Stn. 9 in July and September; and in Stn. 10 in September.

C. On Fecal Coliform: (In Geomean)

Laguna Lake:

- Out of fifteen (15) monitoring stations for this quarter, the geomean counts in the following lake monitoring stations **constantly** conformed with the WQG for Fecal Coliform:
  - Class A - Five (5) lake monitoring stations, namely; Stns. II, IV, V, VIII, and XVI.
  - Class B – One (1) station: Stn. XVII (Fish Sanctuary).
- The highest fecal coliform counts in geomean was 1,010 MPN/100mL recorded in Stn. XX (GEMS) in July while the lowest was both in September in Stns. VIII and XVIII.
- Stn. XX (GEMS) had the highest fecal coliform counts among all the monitoring stations from July to September. Also, the station displayed the poorest water quality in terms of geomean counts among all the stations and one possible reason is its proximity to the shore which is heavily affected by run-off and domestic wastes.

Tributary Rivers:

- The fecal coliform in geomean counts in all monitored tributary river stations always failed to meet the WQG for Classes A to D.
- The highest geomean fecal coliform counts at 62,698,992 MPN/100mL was measured in Stn. 30 (Binangonan River) in July while the lowest was at 3,338 MPN/100mL in Stn. 17U (Pangil River – Upstream) in September.

D. On pH:

Laguna Lake:

- The following lake stations **constantly** conformed with the WQG for pH:
  - Classes A and B – Nine (9) out of the fifteen (15) lake stations, namely; Stns. I, II, IV, VIII, XV, XVI, XVIII, XX and XXIII.
- The monthly pH values ranged from 7.7 to 8.8.
- The highest was registered in Stns. XIII (Taytay) while the lowest was in Stn. XXIII (Lumban) and were both recorded in July.

Tributary Rivers:

- For this quarter:
  - Class A and B – All monitored tributary stations except for Stn. 5 (Tunasan River) which was assessed at Class C in July.
- The pH levels in all monitored stations ranged from 6.7 to 8.6.
- The lowest was noted in Stn. 18 (Siniloan River) in July to September and in Stn. 23 (Baras River) in July while the highest was recorded in Stn. 5 (Tunasan River) in July.

E. On Ammonia:

Laguna Lake:

- For this quarter, the following lake monitoring stations were **consistently** assessed as:
  - Classes A, B and C - All monitoring stations except for Stn. I (Central West Bay) and Stn. II (East Bay) which had months that conformed with Class D.
- The highest ammonia concentration at 0.10 mg/L was registered in Stn. I (Central West Bay) while the lowest at <0.01 mg/L was noted in most of the stations from July to September.

Tributary Rivers:

- Out of the thirty-six (36) monitored tributary river stations, the following were **constantly** assessed as:
  - Classes A, B and C - Two (2) stations, namely; Stn. 17U (Pangil River – Upstream) and Stn. 22B (Tanay River)
  - Class D - Three (3) stations, specifically; Stn. 12 (Molawin Creek), Stn. 18 (Siniloan River) and Stn. 21 (Pililla River)
  - Failed the WQG for Classes A to D – Stns. 2, 3, 4, 5, 6, 7, 8, 8M, 8U, 9, 11, 24, 24U, 26, 27, 28, and 30.
- The monthly ammonia concentrations ranged from <0.01 mg/L to 27.75 mg/L.
- The lowest was reported in Stns. 10 and 17U in July, and Stns. 17U and 19U in August. On the other hand, the highest was in Stn. 3 (Buli Creek).

F. Nitrate:

Laguna Lake:

- All monitored lake stations conformed with the WQG for Classes A, B and C.
- The highest concentration of nitrate was at 1.30 mg/L in Stn. XIX (Muntinlupa) in August while the lowest value <0.01 mg/L was noted in all the fifteen (15) stations in July, in Stns. XXI and XXIII in August, and in Stn. XXIII (Lumban) in September.

Tributary Rivers:

- The monthly nitrate concentrations in all monitored tributary stations conformed with WQG for Classes A, B and C.
- The levels of nitrate ranged from <0.1 mg/L to 4.00 mg/L.
- The lowest concentration was measured in Stns. 3, 4, 8, 8M, 9, and 17U in July, and Stn. 8 and Stn. 4 in August and September, respectively.
- The highest measured nitrate concentration was reported in Stn. 11 (San Juan River).

G. On Inorganic Phosphate:

Laguna Lake:

- For this quarter, the monthly concentrations of inorganic phosphate in fifteen (15) lake monitoring stations ranged from 0.060 mg/L to 0.230 mg/L, thus, all lake stations consistently failed to conform with the WQG for Classes A to D.
- The highest concentration was noted in Stn. IV (Central Bay) in July while the lowest concentration was recorded in Stn. XVIII (Pagsanjan – East Bay) also in July.

Tributary Rivers:

- The inorganic phosphate concentrations in all monitored tributary river stations exceeded 0.05 mg/L, hence, failed to meet the WQG set for Classes A to D except Stns. 19 (Sta. Maria River – Downstream), 22B (Tanay River – Midstream), and 22C (Daranak Falls – Upstream) in July which all had a concentration of 0.05 mg/L hence, conformed with the WQG for Class D.
- The highest concentration at 4.70 mg/L was in Stn. 8 (Sta. Rosa River – Downstream) in September while the lowest was at 0.05 mg/L.

H. Total Suspended Solids:

Laguna Lake:

- For this quarter, the following **consistently** conformed with the WQG set for total suspended solids:
  - Class A – Six (6) lake stations, namely; Stns. II, IV, XV, XVI, XXI, and XXII.
- In July and September, the TSS levels in all monitoring lake stations conformed with Class A except for Stn. VIII (South Bay) in July which fell under Class B. On the other hand, the remaining lake stations in August were occasionally assessed at Classes B, C, and D.
- The highest TSS level was at 91 mg/L in Stn. V (Northern West Bay) in August while the lowest was 3 mg/L in Stn. XIII (Taytay) in September.

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